

Quinn Mair

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EDUCATION

Duke University Pratt School of Engineering

BSE in Electrical and Computer Engineering, Minor in Cultural Anthropology.

- Relevant Coursework: Digital Systems, Signal Processing, Product Design, Entrepreneurship

Durham, North Carolina

Expected May 2027

George Washington High School

International Baccalaureate Diploma. Valedictorian.

Denver, Colorado

Graduated May 2023

EXPERIENCE

Automation Systems/EE Design Intern

HDR

June 2026 – August 2026

Denver, CO

- Drafted and reviewed instrumentation drawings and electrical one-line diagrams for water treatment plant control systems
- Developed an Excel calculator for a client to determine bus overhaul facility footprint size based on bus input parameters
- Programmed and validated PLC ladder logic using Do-more Designer for industrial automation applications
- Supported multidisciplinary design reviews involving controls, electrical, and instrumentation systems

On-Site Construction Manager

Engineers In Action

May 2025 – August 2025

Shiselweni, Eswatini

- Coordinated daily activities of 40+ workers to deliver the longest footbridge in Eswatini (128m)
- Communicated project progress and technical decisions in Siswati during construction and inauguration
- Drafted and iterated drawings using field data and AutoCAD
- First Duke student awarded Emerging Leader of the Year at the annual EIA conference

President

Duke Engineers for International Development

January 2026 – Present

Durham, NC

- Lead planning, budgeting, implementation, and documentation of 4 infrastructure projects across 3 continents
- Fundraise and allocate \$70,000+ annual budget; personally raised ~\$100,000 over 3 years

Co-Founder

Lunch Bunch LLC

May 2024 – September 2025

Durham, NC

- Founded a lunch tray startup: researched and developed an interlocking lunch tray for local schools
- Drafted and submitted a nonprovisional patent, conducted prior art search, and responded to office actions
- Awarded CFCI GoBig Grant and Duke I&E support grant (x2); featured in Duke Startup Showcase (top 18 of 93)

Teaching Assistant, EGR 105

Duke University

August 2025 – December 2025

Durham, NC

- Instructed 200+ students in recursion, algorithm development, Python syntax, and computational methods
- Evaluated student work and collaborated with instructional staff to refine course policies

TPX Product Experience Intern

Comcast

May 2022 – July 2022

Denver, CO

- Designed an internal data management site and updated UI for software engineering teams
- Helped troubleshoot GitHub and terminal issues across company devices

PROJECTS

32-Bit Pipelined Processor + Video Game

- Designed and verified a 32-bit pipelined processor in Verilog implementing hazard detection, forwarding, carry-lookahead arithmetic, and Modified Booth multiplication
- Debugged processor using GTKWave, Vivado, and Python before FPGA deployment to support a custom MIPS video game

Pump Station Control System

- Designed and programmed an industrial control system for a finished water pump station featuring 4 booster pumps, a chlorine injection system, and customizable alarm and trend navigation
- Includes 22-screen HMI with customizable setpoints, alarm settings, and active status updates
- Factory acceptance tested the system using custom PLC and I/O rack

Audio Note Recognition Using Fast Fourier Transform

- Developed a MATLAB application using FFT techniques to identify musical notes from complex audio recordings

SKILLS

Technical Skills: Verilog, PSpice, KiCad, PCB Design, AutoCAD, Python, C, MATLAB, Vivado, Microsoft Office, Jira, Trello

Activities: Duke Tour Guides, Theta Tau, DukeEngage, Hoops for Troops